

Introduction

Can the free-flowing human essence of sport exist alongside the rigid, algorithmic evaluations of artificial intelligence? As AI becomes an integral part of officiating systems, athlete training, and talent identification, its influence on sports is expanding rapidly. At this early stage of adoption, it is crucial to examine not only the benefits these systems promise—namely improved accuracy and efficiency—but also the unintended consequences they may introduce.

This report examines how AI affects three key areas of contemporary sport: competitive fairness, athlete mental health, and inclusivity. Although organizations such as the UEFA¹ Champions League have introduced transparency measures—for example, by having referees briefly explain Video Assistant Referee (VAR) decisions—these efforts remain limited. As a result, skepticism persists among fans, players, and analysts who fear that AI may erode the emotional unpredictability and human spirit that make sport meaningful. Understanding these impacts is necessary for developing solutions that balance technological advancement with the core values of athletic competition.

Background

The current situation in sports reflects a growing skepticism among fans toward AI's integration. A 2024 Ipsos survey found that half of English football fans 50% believe VAR has had a negative impact on the live match-going experience, while 44% think it has harmed football overall (Pedley et al., 2025). Many believe that AI systems, such as the Video Assistant Referee (VAR) in soccer, make game-changing decisions without providing

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UEFA, the *Union of European Football Associations*, is the governing body for football in Europe. It oversees the regulation of the sport across its member nations and organizes major international competitions, including the UEFA Champions League, UEFA Europa League, and the UEFA European Championship for both men's and women's teams.

sufficient transparency about how those decisions are made. Unlike traditional video replay, VAR relies on AI-driven computer-vision models that track players' limbs, generate three-dimensional reconstructions of movement, and calculate offside lines with millimeter precision. Because these calculations occur behind the scenes, fans often perceive VAR as a "black box," raising an important question in the modern sports world: *Should such slim, machine-detected margins decide the outcome of a match* (Farajpour, Amerinia, & Pourjavaheri, 2025)?

A well known example occurred during the FIFA World Cup Final, the most prestigious match in global soccer. The World Cup Final carries extraordinary significance: it is watched by hundreds of millions of viewers worldwide, and its outcome can define a player's career, shape a nation's legacy, and reach audiences who do not follow sports regularly. In the 2022 Final a defender claimed that an opposing striker was offside—a decision that was nearly impossible to see with the naked eye. However, VAR showed that Argentina's Lautaro Martínez was onside by a fraction of a legally playable body part², resulting in a goal that helped secure Argentina's victory. This incident illustrates how deeply AI has become embedded in the sport and how its precision can both clarify and complicate perceptions of fairness.

Beyond soccer, AI technologies have expanded rapidly across major North American sports. In the NBA, advanced tracking systems such as Second Spectrum analyze player movement, spacing, and shooting efficiency in real time. In the NFL, AI is increasingly used for injury prediction, play-pattern recognition, and performance modeling. Major League Baseball (MLB) is also experimenting with automated strike-zone systems designed to assist or eventually replace human umpires. These developments demonstrate that AI is

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Offside Rule: A player is considered offside if any part of the body that can legally play the ball (such as the foot, leg, torso, or head) is closer to the opponent's goal line than the second-last defender at the moment the ball is passed.

influencing entire leagues, shaping how teams train, strategize, and make high-stakes decisions—not just isolated officiating moments (theScore Staff, 2024).

However, access to these technologies remains uneven. Wealthier clubs can invest in sophisticated analytics platforms, high-resolution tracking systems, and specialized AI staff, while smaller organizations struggle to afford even basic data-collection tools. Research from the Barça Innovation Hub indicates that teams with dedicated analytics departments experience measurable advantages in recruitment efficiency and injury prevention—benefits that can translate into competitive success over a season (González, 2025). At the same time, constant biometric monitoring and AI-driven performance evaluation introduce new psychological pressures for athletes, raising concerns about privacy, mental health, and long-term emotional well-being. These trends demonstrate that although AI promises greater accuracy and efficiency, it also produces unintended inequalities and ethical challenges that require careful examination.

Analysis

Fairness and Transparency in AI Officiating (VAR)

Artificial intelligence has become deeply embedded in sports officiating through systems such as the Video Assistant Referee (VAR) and its supporting tools, including semi-automated offside technology (SAOT). VAR is the overarching review system used by human officials, while SAOT is an AI-driven subsystem that tracks player and ball movement to assist with offside decisions. These tools were introduced to improve the speed and accuracy of officiating, and in many cases they have succeeded. According to a FIFA media release, semi-automated offside technology “helps referees make faster, more accurate and more consistent decisions, reducing subjectivity and human error” (FIFA, 2022). This

suggests that AI-based systems can enhance the technical precision of officiating, particularly in situations where margins are too narrow for the human eye to judge reliably.

However, accuracy alone does not guarantee fairness. Unlike standard video replay, VAR relies on AI-driven computer-vision models that track player limbs, reconstruct three-dimensional movement patterns, and calculate offside lines with millimeter precision. In elite competitions, a sensor inside the match ball transmits its position up to 500 times per second, providing additional spatial data that the AI processes in real time. While this advanced infrastructure improves detection consistency, it also makes the officiating process more complex and less transparent. For most players, coaches, and fans, the decision-making logic remains hidden, contributing to what researchers describe as a “black box” system (Farajpour, Amerinia, & Pourjavaheri, 2025).

This opacity becomes especially problematic when high-stakes decisions shape match outcomes. For example, during a 2020 English Premier League match, VAR incorrectly disallowed two legitimate goals, prompting debate about accountability—should responsibility fall on the on-field referee who accepts the VAR communication, the VAR officials interpreting the system output, the software engineers who developed the detection models, or the governing body that approved the technology (Farajpour et al., 2025)? Across multiple leagues, teams that have requested explanations following controversial reviews often receive highly technical or incomplete responses, reinforcing the perception that AI officiating is opaque and unaccountable.

These challenges show that while AI may increase accuracy, it can still undermine fairness when its processes are not clearly communicated. For many fans, fairness is not only about the correctness of the final call but also about understanding how and why that call was made. Without transparent reasoning or consistent post-decision explanations, AI officiating

risks creating greater distrust—even when the technology is technically more accurate than human officials.

Inequality in Access to AI Tools

As AI becomes more embedded in professional athletics, its adoption has amplified existing inequalities between wealthy and financially limited organizations. Top clubs can invest in advanced analytics platforms, high-resolution motion-tracking systems, biomechanical wearables, and AI-driven scouting networks, while smaller clubs often lack the resources or infrastructure to access comparable technologies. Industry trends reflect this widening divide: the global sports analytics market was valued at USD 4.47 billion in 2024 and is projected to exceed USD 14 billion by 2030 (Grand View Research, 2025; Mordor Intelligence, 2025). Similarly, Sports Business Journal (2025) noted that USD 51.9 billion was invested in sports technology in the first half of 2025 alone, with the majority directed toward elite leagues and top-tier clubs.

These financial disparities translate directly into competitive advantages. Clubs with access to AI tools benefit from a variety of mechanisms that enhance their on-field success and financial performance. AI-driven scouting systems analyze thousands of players across global leagues, identifying undervalued talent through pattern-recognition models that outperform traditional methods. Tactical analytics platforms, such as Second Spectrum, process real-time movement data to inform strategic decisions. AI-powered dynamic pricing systems optimize ticket revenue, while injury-prediction algorithms help maintain player availability throughout the season. Together, these systems increase a club's competitive performance, reduce costs, and generate higher revenue (IJBMI, 2025), creating greater capacity for reinvestment in technology.

A clear real-world example is Liverpool FC, widely regarded as one of the earliest and most successful adopters of data-driven scouting in European football. Through a recruitment approach heavily informed by AI-enhanced analytics, Liverpool identified undervalued players such as Mohamed Salah, Sadio Mané, Roberto Firmino, and Andrew Robertson—all acquired for modest transfer fees compared to market standards. These players became core contributors to Liverpool's resurgence, leading to the 2019 UEFA Champions League title and the club's first Premier League championship in 30 years in 2020. Analysts and club insiders credit this success to Liverpool's machine-learning scouting models and predictive performance algorithms, which identified attributes other clubs had overlooked (Molloy, 2020; Rudd, 2019). This example shows how AI can dramatically elevate a club's performance and financial stability—and also how teams without access to similar systems cannot replicate these advantages, further widening the competitive gap.

As AI increasingly shapes recruitment, strategy, and player development, technology-rich organizations strengthen their advantages season after season. This creates a reinforcing feedback loop, where success generates revenue, which funds further technological investment, widening the gap even more. Over time, this cycle risks undermining the principle of fair and balanced competition that underpins modern sport.

Mental Health and Athlete Well-Being

The integration of AI into athlete monitoring has introduced sophisticated tools for optimizing performance, but it has also created new psychological pressures. Contemporary systems track a wide range of physiological indicators—such as heart-rate variability, sleep quality, stress biomarkers, gait patterns, and micro-movements—to refine training loads and identify early signs of fatigue. A recent study demonstrated that an AI-based mental-state framework could detect sub-clinical stress 6.7 minutes before athletes became consciously aware of it,

and reduce the duration of acute stress episodes by approximately 42% (Badicu & Silva, 2025). While these capabilities offer meaningful advantages for performance management, they also intensify concerns regarding constant surveillance, privacy, and mental burden.

For many athletes, the knowledge that every biometric fluctuation is continuously recorded and analyzed can create heightened anxiety. Negative deviations—such as elevated stress levels or declining recovery metrics—may be interpreted not only as performance concerns but as potential risks for reduced playing time, contract renegotiation, or loss of competitive standing. This dynamic can generate a self-reinforcing cycle of stress, in which athletes feel pressured to monitor themselves obsessively, attempt to “game” biometric systems, or suppress symptoms to avoid appearing unfit.

Furthermore, a growing body of research on digital mental-health tools in professional sport indicates that continuous monitoring can unintentionally increase athletes’ vulnerability to stress when performance data becomes intertwined with organizational decision-making. In many elite leagues, teams now use wearable systems and biometric platforms—such as GPS trackers, sleep-monitoring devices, and real-time stress-detection dashboards—to evaluate players’ physical and psychological readiness. Although these technologies are designed to support performance, athletes often report feeling judged by the numbers rather than by holistic coaching feedback. This can create pressure to maintain “perfect metrics” and heighten anxiety about how coaches, analysts, or contract evaluators interpret their data. As a result, the same AI systems intended to enhance performance can also amplify psychological strain, blurring the line between helpful monitoring and intrusive surveillance.

Ethical and Cultural Implications

Beyond technical and financial considerations, AI’s influence raises broader ethical and cultural questions about the evolving nature of sport. As algorithms take on a growing role in

officiating, training, and strategic decision-making, they risk displacing the intuition, spontaneity, and creativity that traditionally define athletic performance. Scholars such as Millington argue that this shift threatens the “human spirit” of competition, transforming dynamic contests into data-optimized performances governed by efficiency metrics rather than human expression. This creates a deeper philosophical dilemma: *should technological precision be pursued even if it diminishes the imperfections, emotional unpredictability, and spontaneous moments that make sport meaningful to both athletes and fans?*

AI adoption may also contribute to the homogenization of global sports culture. Historically, distinct playing styles reflected national identity—such as Brazil’s improvisational flair in football, Spain’s possession-heavy passing (“tiki-taka”), or Japan’s disciplined approach to baseball. As analytics-driven optimization grows more dominant, teams are increasingly encouraged to employ similar structures, risk-minimization strategies, and performance models derived from the same data-driven logic. This can reduce stylistic diversity on the field, narrowing the expressive range that once differentiated sporting cultures (Carrio Sampedro, 2023).

Furthermore, the cultural symbolism of sport—its role in expressing community identity, national pride, and collective memory—may be reshaped as decisions shift from human judgment to algorithmic assessment. When algorithms become central to determining tactics, evaluating talent, and making officiating calls, they also influence what fans perceive as legitimate, exciting, or “authentic” competition. In this sense, AI is transforming not only *how* sports are played but also *what they represent*: a global cultural practice rooted in uncertainty, shared emotion, and human connection.

Together, these ethical and cultural challenges reveal that the consequences of AI extend far beyond performance metrics. While AI offers substantial benefits, it also introduces complex questions about fairness, identity, meaning, and the future character of sport—questions that

must be addressed thoughtfully as technology becomes further integrated into athletic competition.

Solutions

Fairness and Transparency

Although AI officiating systems such as VAR have demonstrated high levels of accuracy—particularly in situations where human officials face physical or perceptual limitations—their lack of transparency remains a major barrier to public trust. The core problem is not the correctness of AI decisions, but how these decisions are perceived. When algorithms operate behind the scenes and their reasoning is not communicated, fans and players often view AI as intrusive and unaccountable. Strengthening fairness therefore requires solutions that prioritize both transparency and clear responsibility.

One effective approach is to implement transparent documentation requirements, similar to the regulatory model used in California’s legislation on facial recognition technology. In that case, companies were required to provide accessible documentation explaining how their algorithms worked before the technology could be adopted by law enforcement agencies (Mensah, 2023). A similar baseline standard in sport would require vendors to disclose what the system measures, how decisions are produced, and what error margins or model limitations exist. Importantly, this framework preserves trade secrets while establishing shared understanding and traceability, allowing leagues and referees to know *who is accountable* when errors occur.

Additionally, incorporating explainable AI (XAI) into officiating systems would make decisions more interpretable to non-experts. XAI models generate simplified explanations that highlight which data points triggered a call, enabling fans, coaches, and officials to understand why a ruling was made. Broadcasting these explanations or integrating them into referee announcements would significantly improve perceptions of fairness and reduce distrust.

Accountability can also be reinforced by requiring multi-level oversight:

- Vendors ensure proper algorithm documentation and testing.
- Leagues maintain protocols for reviewing contested decisions.
- Referees and officials receive training on the limitations of the system.

This layered model establishes responsibility across all stakeholders rather than allowing blame to fall ambiguously on “the technology.”

Finally, collaborative practices—such as shared audit panels, cross-league review groups, or open-source data initiatives—can increase transparency and independent oversight without forcing vendors to reveal proprietary code. These collaborative models give governing bodies and referees a role in evaluating AI behavior, ensuring that officiating remains accountable to the sport rather than solely to technology providers.

Together, these solutions address the core issue: the opacity of AI decision-making. By improving transparency, integrating explainable AI, and clarifying accountability roles, leagues can strengthen public trust and ensure AI enhances fairness rather than appearing to undermine it.

Inequality in Access to AI Tools

Before proposing solutions, it is important to acknowledge a fundamental reality: professional sport—like any competitive industry—will always contain performance disparities. Financial differences, market size, and organizational strategy naturally produce uneven outcomes. However, when AI technologies amplify these disparities to the point

where lower-resourced teams cannot realistically compete, the integrity and unpredictability of sport are threatened. Therefore, solutions should aim not to eliminate competitive edges entirely, but to prevent technological inequality from becoming structurally entrenched.

A practical first step is to establish a baseline level of AI infrastructure that all teams within a league must have access to. This could include standardized performance-tracking systems, basic analytics platforms, and foundational scouting tools funded collectively through league-wide contributions. This model mirrors existing cost-sharing arrangements seen in leagues such as the NBA, where all franchises contribute to shared expenses related to officiating, broadcasting technology, and league operations. By ensuring that every team has access to essential AI capabilities, leagues can maintain a minimum standard of competitiveness while still allowing wealthier clubs to pursue additional innovations.

To complement this, leagues could implement AI spending regulations similar in spirit to the NBA's salary-cap structure. Rather than restricting talent acquisition, these policies would establish limits on the amount clubs can invest in proprietary AI systems, data infrastructures, or exclusive analytics partnerships within a given season. Much like the NBA's "first apron"³ and "second apron"⁴ luxury-tax thresholds designed to prevent top teams from hoarding elite players, AI investment caps would prevent elite teams from monopolizing cutting-edge technologies. Importantly, these regulations would not prohibit top clubs from accessing advanced tools; instead, they would prevent disproportionate stockpiling that widens the performance gap and locks smaller clubs into permanent disadvantage.

Together, these strategies help interrupt the feedback loop in which wealthy clubs acquire better AI systems, gain competitive advantages, earn more revenue, and reinvest in even more advanced technologies—leaving lower-resourced teams perpetually behind. By

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A spending threshold in the NBA's salary-cap system. Teams that exceed it face added roster and trade restrictions designed to limit competitive imbalance.

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A higher spending tier with even stricter penalties—such as limits on player acquisition and future draft flexibility—to prevent wealthy teams from stockpiling talent.

providing every team with a foundational toolkit and placing reasonable boundaries on excessive technological spending, leagues can protect the competitive balance and unpredictability that define sport as a compelling form of entertainment.

Mental Health and Athlete Well-Being

To address the psychological burdens created by continuous biometric monitoring, AI systems need to shift from rigid, performance-only evaluation to more context-sensitive and athlete-centered frameworks (Mensah, 2023). Currently, many AI tools interpret biometric fluctuations—such as elevated stress, reduced sleep quality, or inconsistent movement patterns—without considering the human circumstances surrounding those metrics. This lack of context can cause athletes to feel misrepresented, over-scrutinized, or unfairly judged.

One solution is to integrate context-aware evaluation into monitoring systems. This approach combines biometric data with additional information about an athlete's mental state, perceived exertion, and off-field factors that may affect performance. Instead of interpreting a spike in stress as a singular failure, the system would contextualize it with self-reported mood, fatigue, personal circumstances, or recent scheduling demands. This reframing transforms AI metrics from rigid judgments into a more holistic representation of athlete well-being, helping to reduce anxiety associated with “perfect metric” expectations.

A second solution is to give athletes greater agency and ownership over how AI systems are used in their training environment. Allowing athletes to view their own data first, annotate unusual readings, or request temporary pauses in monitoring encourages them to participate in the evaluation process rather than feel subjected to it. This reduces feelings of

surveillance and shifts responsibility from being externally imposed by coaches or management to being jointly managed by athletes and staff.

Finally, teams can adopt dual-layered feedback processes in which biometric readings are reviewed alongside human input from sports psychologists, medical staff, and athletes themselves. This blended approach ensures that data interpretations are guided by professional expertise rather than solely by algorithmic assessment. When AI insights are combined with athlete-centered feedback, monitoring becomes more supportive than punitive, preserving performance benefits while reducing the mental strain associated with continuous evaluation.

Together, these solutions help mitigate the psychological pressure of AI-based monitoring by reframing how data is gathered, interpreted, and communicated. By incorporating context, prioritizing athlete autonomy, and embedding human oversight, teams can retain the performance advantages of AI while ensuring systems promote — rather than compromise — athlete mental well-being.

Ethical and Cultural Implications

To address concerns that AI may erode the human spirit of competition, solutions must focus on integrating technology in ways that enhance—rather than replace—human creativity, intuition, and identity. One approach is to design AI systems that intentionally incorporate human-centered parameters rather than optimizing solely for efficiency. For example, instead of generating only the mathematically optimal tactical decision, AI tools could be programmed to factor in coaching philosophy, team identity, and stylistic preferences. This allows AI to complement, rather than overshadow, the strategic intuition that defines sport.

To prevent the homogenization of playing styles, leagues and technology providers can adopt context-aware tactical algorithms. These systems would tailor their recommendations to a team's historical identity, preferred style of play, roster characteristics, and cultural expectations of their fan base. In practice, this means AI tools would help teams strengthen their unique approach—whether that is high-pressing football, possession play, counterattacking strategies, or regionally distinctive techniques—while still providing evidence-based insights for competitive improvement. This preserves stylistic diversity while ensuring that performance optimization does not erase cultural expression.

The cultural symbolism of sport can also be protected by ensuring that AI-based decision-making remains tightly connected to the emotional and communal aspects of fan experience. AI systems used in broadcasting, analytics, or fan engagement platforms could incorporate fan-context data—such as long-term attachment to specific styles, rivalry traditions, or regional identities—to guide how narratives and insights are delivered. This ensures AI does not flatten sporting culture into a uniform analytic product but instead strengthens the connection between fans and their teams.

Finally, because many ethical concerns overlap with issues of privacy, transparency, and autonomy, any culturally sensitive AI system must include strict safeguards for voluntary, athlete-consented data sharing and clearly defined limits on what personal information can be collected or analyzed (González, 2025). These boundaries ensure that efforts to preserve cultural and human dimensions do not create new forms of intrusion or surveillance.

Together, these strategies position AI as a tool that enhances performance and entertainment value without compromising the traditions, identities, and emotional richness that make sport meaningful to athletes and fans alike.

Vision for the Future

Looking ahead, the role of AI in sport will depend on how effectively leagues, governing bodies, and organizations implement the solutions discussed in this report. If transparency frameworks, athlete-centered monitoring practices, and equitable access standards become widely adopted, AI has the potential to strengthen sport rather than undermine it. In this future, officiating systems would provide clear and explainable decisions, players would benefit from performance tools that respect their mental and emotional well-being, and teams across all competitive levels would have access to a baseline level of AI infrastructure.

As AI technologies mature, they will likely become more collaborative in design—integrating human judgment, coaching philosophy, cultural identity, and fan experience into their recommendations. Instead of replacing human intuition, AI could evolve into a support system that enhances decision-making while preserving the unpredictability, diversity, and creativity that define sport. At the same time, leagues may develop clearer governance models that establish accountability, protect athlete data, and ensure that AI is used in ways that align with the values of fairness, inclusivity, and cultural expression.

If these improvements continue, the future of AI in sport will not be one of domination or displacement, but of balanced integration—where technology strengthens competition, safeguards well-being, and enhances the authenticity of athletic performance. Ultimately, the trajectory of AI in sport will depend not on the technology itself, but on the governance decisions, ethical standards, and cultural priorities that shape its implementation.

Conclusion

The rapid integration of artificial intelligence into professional sport has created both promising opportunities and complex challenges. While AI enhances accuracy in officiating, expands analytical capabilities, and optimizes athlete performance, it also raises concerns related to transparency, competitive inequality, mental health, and cultural identity. These issues do not stem from the technology itself, but from how AI systems are implemented, governed, and communicated.

This report has shown that the *free-flowing human essence* of sport can coexist with the rigid, algorithmic evaluations produced by AI—but only when AI is designed to complement, rather than override, human judgment. With explainable systems, culturally aware models, and athlete-centered monitoring practices, AI can support creativity, intuition, and emotional expression instead of diminishing them. In this balanced future, technology enhances the game without defining it. At the same time, sports cannot ignore the core question raised earlier: *should technological precision be pursued even if it risks diminishing the imperfections, unpredictability, and spontaneous moments that make sport meaningful?* The answer lies in using AI as an enhancement tool—not a replacement for the drama and humanity that give sport its emotional power.

Similarly, while slim, machine-detected margins can help ensure that decisions are accurate, they should not determine the outcome of a match in isolation. Transparency frameworks, interpretable decision displays, and clear accountability standards are essential to ensure that such margins are understood, trusted, and evaluated fairly. When AI decisions are clearly communicated and contextualized—rather than hidden behind opaque algorithms—these margins can contribute to fairness without eroding the human spirit of competition.

Ultimately, the future of AI in sport will be determined not by the sophistication of algorithms, but by the ethical and organizational choices made by leagues, developers, and governing bodies. With thoughtful regulation, equitable access, and a commitment to preserving cultural and human values, AI can strengthen sport without compromising the qualities that make it meaningful. Given the scale of AI adoption across global sport, addressing these challenges is essential to ensure that technology enhances rather than diminishes competitive integrity. Ensuring this balance will be crucial as AI continues to shape the next generation of athletic competition.

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